

A Technical Introduction to Curse of dimensionality

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} r^d$$

Section 1

Therefore, the squared distance from the origin, $r^2 = \sum_{i=1}^d x_i^2$ has the average value $d/3$ and variance $4d/45$. For large d , distribution of r^2/d is close to the normal distribution with the mean $1/3$ and the standard deviation $2/\sqrt{45d}$ according to the central limit theorem. Thus, when uniformly generating points in high dimensions, both the "middle" of the hypercube, and the corners are empty, and all the volume is concentrated near the surface of a sphere of "intermediate" radius $d/3$. This also helps to understand the chi-squared distribution. Indeed, the (non-central) chi-squared distribution associated to a random point in the interval $[-1, 1]$ is the same as the distribution of the length-squared of a random point in the d -cube. By the law of large numbers, this distribution concentrates itself in a narrow band around d times the standard deviation squared (σ^2) of the original derivation. This illuminates the chi-squared distribution and also illustrates that most of the volume of the d -cube concentrates near the boundary of a sphere of radius σd . A further development of this phenomenon is as follows. Any fixed distribution on the real numbers induces a product distribution on points in $\mathbb{R}^d$. For any fixed n , it turns out that the difference between the minimum and the maximum distance between a random reference point Q and a list of n random data points $P_1, \dots, P_n$ become indiscernible compared to the minimum distance:

Section 2

■ x_i^2 ■ = $\int_{-1}^1 x^2 dx = 1/3$ $\left\langle x_i^2 \right\rangle = \frac{1}{2} \int_{-1}^1 x^2 dx = \frac{1}{3}$. The variance of x_i^2 for uniform distribution in the cube is

Section 3

$V_{\text{hypersphere}} V_{\text{hypercube}} = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} \frac{d^d}{2^d} \rightarrow 0$ as $d \rightarrow \infty$. Furthermore, the distance between the center and the corners is $r d$, which increases without bound

for fixed r . In this sense when points are uniformly generated in a high-dimensional hypercube, almost all points are much farther than r units away from the center. In high dimensions, the volume of the d -dimensional unit hypercube (with coordinates of the vertices ± 1) is concentrated near a sphere with the radius $d/3$ for large dimension d . Indeed, for each coordinate x_i the average value of x_i^2 in the cube is

Section 4

Machine learning In machine learning problems that involve learning a "state-of-nature" from a finite number of data samples in a high-dimensional feature space with each feature having a range of possible values, typically an enormous amount of training data is required to ensure that there are several samples with each combination of values. In an abstract sense, as the number of features or dimensions grows, the amount of data we need to generalize accurately grows exponentially. A typical rule of thumb is that there should be at least 5 training examples for each dimension in the representation. In machine learning and insofar as predictive performance is concerned, the curse of dimensionality is used interchangeably with the peaking phenomenon, which is also known as Hughes phenomenon. This phenomenon states that with a fixed number of training samples, the average (expected) predictive power of a classifier or regressor first increases as the number of dimensions or features used is increased but beyond a certain dimensionality it starts deteriorating instead of improving steadily. Nevertheless, in the context of a simple classifier (e.g., linear discriminant analysis in the multivariate Gaussian model under the assumption of a common known covariance matrix), Zollanvari et al. showed both analytically and empirically that as long as the relative cumulative efficacy of an additional feature set (with respect to features that are already part of the classifier) is greater (or less) than the size of this additional feature set, the expected error of the classifier constructed using these additional features will be less (or greater) than the expected error of the classifier constructed without them. In other words, both the size of additional features and their (relative) cumulative discriminatory effect are important in observing a decrease or increase in the average predictive power. In metric learning, higher dimensions can sometimes allow a model to achieve better performance. After normalizing embeddings to the surface of a hypersphere, FaceNet achieves the best performance using 128 dimensions as opposed to 64, 256, or 512 dimensions in one ablation study. A loss function for unitary-invariant dissimilarity between word embeddings was found to be minimized in high dimensions.

Section 5

A common practice of data mining in this domain would be to create association rules between genetic mutations that lead to the development of cancers. To do this, one would have to loop through each genetic mutation of each individual and find other genetic mutations that occur over a desired threshold and create pairs. They would start with pairs of two, then three, then four until they result in an empty set of pairs. The complexity of this algorithm can lead to calculating all permutations of gene pairs for each individual or row. Given the formula for calculating the permutations of n items with a group size of r is: $\frac{n!}{(n-r)!}$, calculating the number of three pair permutations of any given individual would be 7988004000 different pairs of genes to evaluate for each individual. The number of pairs created will grow by an order of factorial as the size of the pairs increase. The growth is depicted in the permutation table (see right). As we can see from the permutation table above, one of the major problems data miners face regarding the curse of dimensionality is that the space of possible parameter values grows exponentially or factorially as the number of features in the data set grows. This problem critically affects both computational time and space when searching for associations or optimal features to consider. Another problem data miners may face when dealing with too many features is that the number of false predictions or classifications tends to increase as the number of features grows in the data set. In terms of the classification problem discussed above, keeping every data point could lead to a higher number of false positives and false negatives in the model. This may seem counterintuitive, but consider the genetic mutation table from above, depicting all genetic mutations for each individual. Each genetic mutation, whether they correlate with cancer or not, will have some input or weight in the model that guides the decision-making process of the algorithm. There may be mutations that are outliers or ones that dominate the overall distribution of genetic mutations when in fact they do not correlate with cancer. These features may be working against one's model, making it more difficult to obtain optimal results. This problem is up to the data miner to solve, and there is no universal solution. The first step any data miner should take is to explore the data, in an attempt to gain an understanding of how it can be used to solve the problem. One must first understand what the data means, and what they are trying to discover before they can decide if anything must be removed from the data set. Then they can create or use a feature selection or dimensionality reduction algorithm to remove samples or features from the data set if they deem it necessary. One example of such methods is the interquartile range method, used to remove outliers in a data set by calculating the standard deviation of a feature or occurrence.